전산화학 (2011년 1학기) **1차 시험 해답**

(총 5 문제, 70 점)

1. **(총 10점, 각 5점)**

(a)

$$\begin{aligned} z &= (1+\sqrt{2}i)e^{-\pi/3} \\ &= (1+\sqrt{2}i)\Big(\cos\Big(-\frac{\pi}{3}\Big)+i\sin\Big(-\frac{\pi}{3}\Big)\Big) \\ &= (1+\sqrt{2}i)\Big(\cos\Big(\frac{\pi}{3}\Big)-i\sin\Big(\frac{\pi}{3}\Big)\Big) \\ &= (1+\sqrt{2}i)\Big(\frac{1}{2}-\frac{\sqrt{3}}{2}i\Big) \\ &= \frac{1}{2}-\frac{\sqrt{3}}{2}i+\frac{\sqrt{2}}{2}i+\frac{\sqrt{6}}{2} \\ &= \Big(\frac{1}{2}+\frac{\sqrt{6}}{2}\Big)+\Big(-\frac{\sqrt{3}}{2}+\frac{\sqrt{2}}{2}\Big)i \end{aligned}$$

그러므로 $\mathrm{Re}(z)=\frac{1}{2}+\frac{\sqrt{6}}{2}$ 이고, $\mathrm{Im}(z)=-\frac{\sqrt{3}}{2}+\frac{\sqrt{2}}{2}$ 이다.

(b)

$$\begin{aligned} z=\Big(-\frac{\sqrt{3}}{2}+\frac{1}{2}i\Big)^{122} &= \Big(\cos\Big(\frac{5\pi}{6}\Big)+i\sin\Big(\frac{5\pi}{6}\Big)\Big)^{122} \\ &= \cos\Big(122\times\frac{5\pi}{6}\Big)+i\sin\Big(122\times\frac{5\pi}{6}\Big) \qquad \text{(De Moivre 공식에 의해)} \\ &= \cos\Big(61\times\frac{5\pi}{3}\Big)+i\sin\Big(61\times\frac{5\pi}{3}\Big) \\ &= \cos\Big(60\times\frac{5\pi}{3}+\frac{5\pi}{3}\Big)+i\sin\Big(60\times\frac{5\pi}{3}+\frac{5\pi}{3}\Big) \\ &= \cos\Big(100\pi+\frac{5\pi}{3}\Big)+i\sin\Big(100\pi+\frac{5\pi}{3}\Big) \\ &= \cos\Big(\frac{5\pi}{3}\Big)+i\sin\Big(\frac{5\pi}{3}\Big) \\ &= \cos\Big(2\pi-\frac{\pi}{3}\Big)+i\sin\Big(2\pi-\frac{\pi}{3}\Big) \\ &= \cos\Big(\frac{\pi}{3}\Big)-i\sin\Big(\frac{\pi}{3}\Big)=\frac{1}{2}-\frac{\sqrt{3}}{2}i \end{aligned}$$

$\therefore \mathrm{Re}(z)=\frac{1}{2}, \qquad \mathrm{Im}(z)=-\frac{\sqrt{3}}{2}.$

2. **(10점)** $-a$ 부터 a 까지 적분했을 때 항상 0이 되기 위해서는 $f(x)$ 는 기함수이어야 한다.

(a) $f(-x)=(e^{-x}-e^{x})/2=-(e^{x}-e^{-x})/2=-f(x)$ 이므로 **기함수이다.**

(b) $f(-x)=e^{-x}\sin(-x)=-e^{-x}\sin(x)\neq -f(x)$ 이므로 **기함수가 아니다.**

(c) $f(-x)=e^{-(-x)^2}\sin(-x)=-e^{-x^2}\sin x=-f(x)$ 이므로 **기함수이다.**

(d) $f(-x)=\cos(-x)+\sin(-x)=\cos x-\sin x\neq -f(x)$ 이므로 **기함수가 아니다.**

그러므로 적분 값이 a 에 관계없이 항상 0이 되는 것은 **(a)와 (c)** 이다.

3. **(총 15점, 각 5점)**

(a) 문제의 (1) 식을 다시 쓰면

$$\begin{aligned}\psi_{211}(r,\theta,\phi) &= \frac{1}{8\sqrt{\pi}}\left(\frac{1}{a_0}\right)^{3/2}\frac{r}{a_0}\sin(\theta)e^{i\phi} \\ &= \frac{1}{8\sqrt{\pi}}\left(\frac{1}{a_0}\right)^{3/2}\frac{r}{a_0}\sin(\theta)\big(\cos(\phi)+i\sin(\phi)\big) \qquad (1)\end{aligned}$$

그러므로

$$\begin{aligned}\mathrm{Re}(\psi_{211}) &= \frac{1}{8\sqrt{\pi}}\left(\frac{1}{a_0}\right)^{3/2}\frac{r}{a_0}\sin(\theta)\cos(\phi), \\ \mathrm{Im}(\psi_{211}) &= \frac{1}{8\sqrt{\pi}}\left(\frac{1}{a_0}\right)^{3/2}\frac{r}{a_0}\sin(\theta)\sin(\phi)\end{aligned}$$

(b)

$$|\psi_{211}(r,\theta,\phi)| = \frac{1}{8\sqrt{\pi}}\left(\frac{1}{a_0}\right)^{3/2}\frac{r}{a_0}\sin(\theta) \qquad (\text{단, } 0\le\theta\le\pi)$$

(c) 3차원 직교좌표와 구면 극좌표의 관계식으로부터

$$x = r\sin\theta\cos\phi$$
$$y = r\sin\theta\sin\phi$$

0이므로, 위의 (1) 식으로부터

$$\begin{aligned}\psi_{211}(r,\theta,\phi) &= \frac{1}{8\sqrt{\pi}}\left(\frac{1}{a_0}\right)^{3/2}\frac{r}{a_0}\sin(\theta)\big(\cos(\phi)+i\sin(\phi)\big) \\ \psi_{211}(x,y,z) &= \frac{1}{8\sqrt{\pi}}\left(\frac{1}{a_0}\right)^{3/2}\frac{1}{a_0}(x+iy)\end{aligned}$$

4. **(총 15점)**

(a) **(5점)**

$$I = \int_{-\infty}^{\infty} dx e^{-a^2x^2} = \int_{-\infty}^{\infty} dy e^{-a^2y2}. \qquad (2)$$

$$\begin{aligned}I^2 &= \int_{-\infty}^{\infty} dx e^{-a^2x^2} \times \int_{-\infty}^{\infty} dy e^{-a^2y^2}, \\ &= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} dxdy e^{-a^2(x^2+y^2)}, \\ &\quad \text{직교 좌표의 극 좌표 변환을 이용하면} \\ &= \int_0^{2\pi}\int_0^{\infty} rdrd\theta e^{-a^2r^2}, \\ &\quad t = ar \text{ 로 치환하면,} \\ &= \frac{1}{a^2}\int_0^{2\pi} d\theta \int_0^{\infty} dtte^{-t^2}, \\ &= \frac{1}{a^2}2\pi\Big[-\frac{1}{2}e^{-t^2}\Big]_0^{\infty} \\ &= \frac{\pi}{a^2}\end{aligned}$$

$$\therefore I = \int_{-\infty}^{\infty} e^{-a^2x^2}dx = \frac{\sqrt{\pi}}{a}. \qquad (3)$$

(b) **(10점)** $a = (1-i)/\sqrt{2}$ 이므로

$$a^2 = \Big(\frac{1-i}{\sqrt{2}}\Big)^2 = \Big(\frac{1-i}{\sqrt{2}}\Big)\Big(\frac{1-i}{\sqrt{2}}\Big) = \frac{1-2i-1}{2} = -i \tag{4}$$

(4) 식을 $\int_{-\infty}^{\infty} e^{-a^2x^2}dx$ 에 넣으면,

$$\int_{-\infty}^{\infty} e^{-a^2x^2}dx = \int_{-\infty}^{\infty} e^{ix^2}dx = \int_{-\infty}^{\infty} \cos x^2 dx + i\int_{-\infty}^{\infty} \sin x^2 dx \tag{5}$$

한편, (a)의 (3) 식으로부터 $\int_{-\infty}^{\infty} e^{-a^2x^2}dx = \sqrt{\pi}/a$ 이므로

$$\frac{\sqrt{\pi}}{a} = \frac{\sqrt{\pi}}{(1-i)/\sqrt{2}} = \frac{\sqrt{2\pi}}{1-i} = \frac{\sqrt{2\pi}}{1-i}\times\frac{1+i}{1+i} = \frac{\sqrt{2\pi}}{2}(1+i) \tag{6}$$

(5) 식과 (6) 식을 비교하고 복소수 상등 정리를 이용하면

$$\begin{aligned}\int_{-\infty}^{\infty} \cos x^2 dx &= \frac{\sqrt{2\pi}}{2}\\ \int_{-\infty}^{\infty} \sin x^2 dx &= \frac{\sqrt{2\pi}}{2}\end{aligned}$$

5. **(총 20점)**

(a) **(5점)** Euler 공식으로부터

$$e^{i\theta} = \cos\theta + i\sin\theta, \tag{7a}$$

$$e^{-i\theta} = \cos\theta - i\sin\theta, \tag{7b}$$

그러므로 (7a) + (7b) 하면

$$\boxed{2\cos\theta = e^{i\theta} + e^{-i\theta} \Rightarrow \cos\theta = \frac{(e^{i\theta}+e^{-i\theta})}{2}} \tag{8}$$

(b) **(5점)**

$$\begin{aligned}\int_0^{\pi} e^{i2n\theta}d\theta &= \Big[\frac{1}{2ni}e^{i2n\theta}\Big]_0^{\pi} \qquad (n = \pm1, \pm2, \pm3, \cdots)\\ &= \frac{1}{2ni}\big(e^{i2n\pi} - 1\big)\\ &= \frac{1}{2ni}\big(\cos(2n\pi) + i\sin(2n\pi) - 1\big)\\ &\quad \cos(2n\pi) = 1 \qquad \text{그리고} \qquad \sin(2n\pi) = 0 \ \text{이므로}\\ &= 0\end{aligned}$$

즉,

$$\boxed{\int_0^{\pi} e^{i2n\theta}d\theta = 0 \qquad (n = \pm1, \pm2, \pm3, \cdots)} \tag{9}$$

(c) **(10점)**

$$\begin{aligned}\int_0^{\pi} \cos^4\theta d\theta &= \int_0^{\pi}\left(\frac{e^{\theta}+e^{-\theta}}{2}\right)^4 \qquad (\text{(a)의 (8) 식으로부터})\\ &= \frac{1}{16}\int_0^{\pi}(e^{2\theta} + 2 + e^{-2\theta})^2 d\theta\\ &= \frac{1}{16}\int_0^{\pi}(e^{4\theta} + 4 + e^{-4\theta} + 4e^{2\theta} + 4e^{-2\theta} + 2)d\theta \qquad (10)\end{aligned}$$

(b)의 (9) 식으로부터

$$\int_0^{\pi} e^{4\theta} d\theta = \int_0^{\pi} e^{-4\theta} d\theta = \int_0^{\pi} e^{2\theta} d\theta = \int_0^{\pi} e^{-2\theta} d\theta = 0 \tag{11}$$

0이므로 (10) 식은

$$\boxed{\int_0^{\pi} \cos^4\theta d\theta = \frac{1}{16}\int_0^{\pi} 6 d\theta = \frac{3}{8}\pi}$$